

Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

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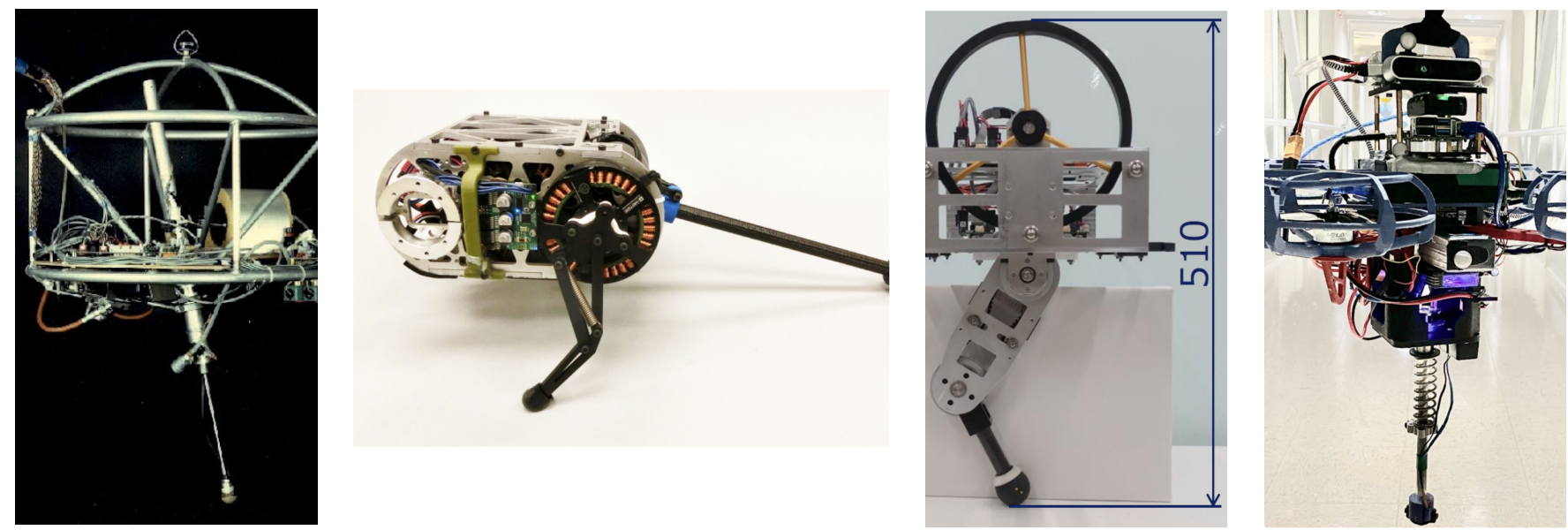
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1. Problem - hybrid dynamics create an authority gap

**Conventional Hopper** → **Inertial Appendages** → **Propeller Assistance**

Stance

Ground reaction force exists only during a short contact interval.

Flight

No contact: attitude becomes underactuated.

Disturbance

Impacts and pushes inject attitude and horizontal momentum.

Control authority disappears exactly when attitude can keep drifting.

Design question: how should leg contact force and rotor thrust be coordinated across stance and flight?

2. First proof - stable on 30° grass



Propellers off

After asymmetric contact, attitude error grows. With no flight-phase actuator, the robot falls.

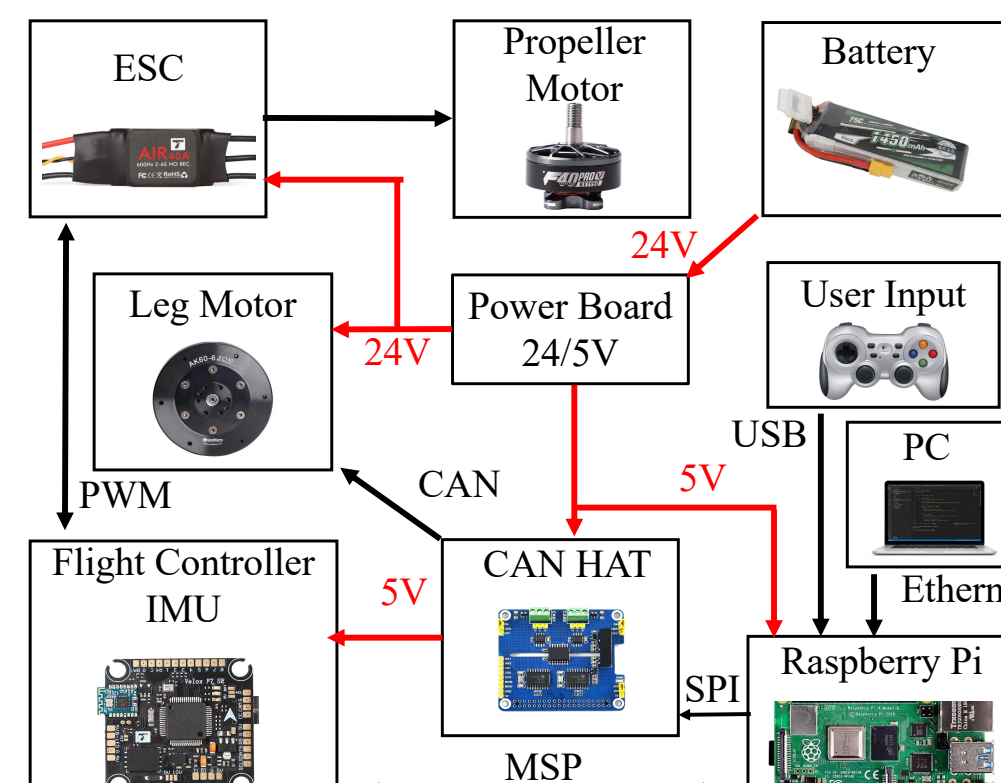
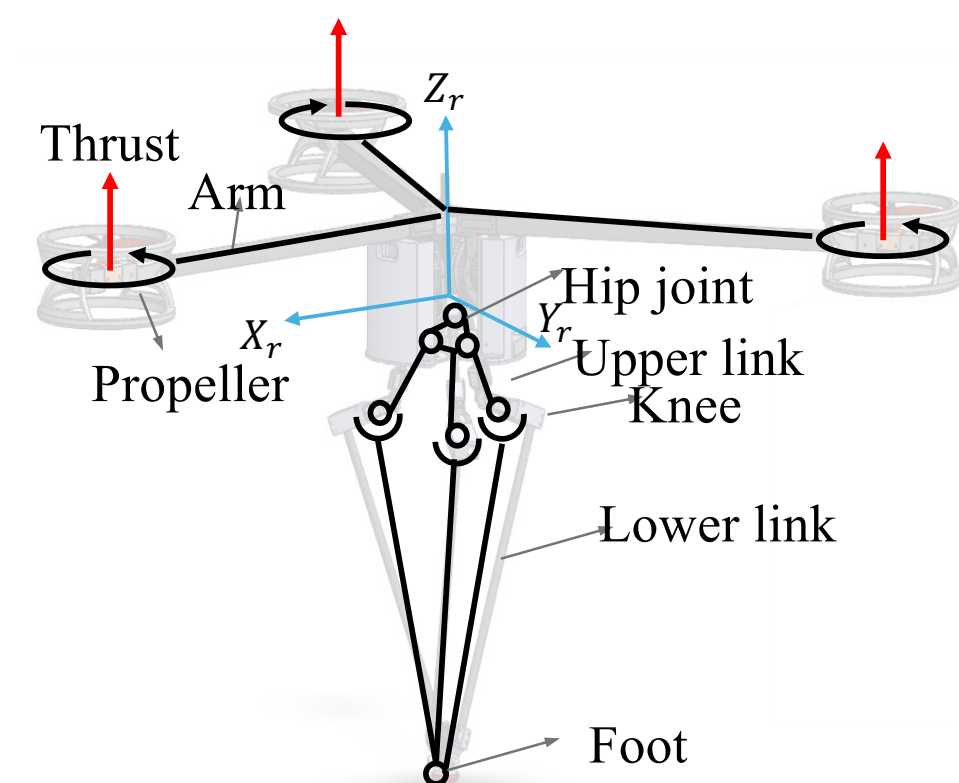
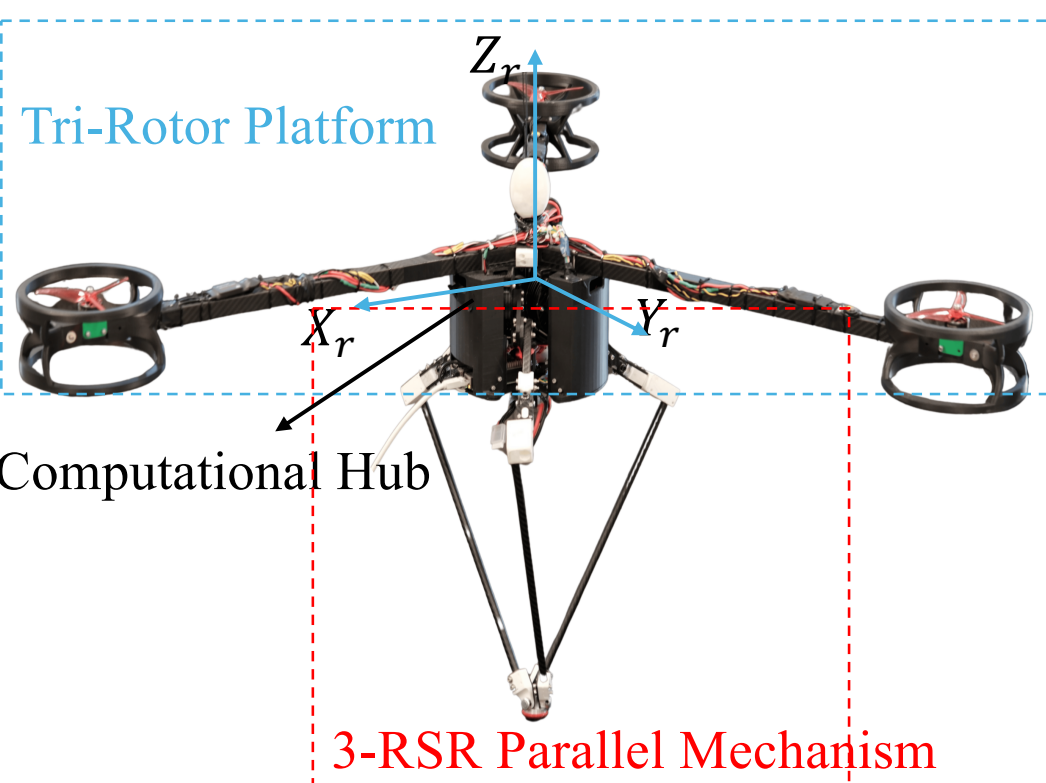
Propellers on

Differential thrust remains available through flight and restores attitude before the next landing.

What this establishes

Residual authority in stance.**Primary** attitude authority in flight.**Faster recovery** from uneven impacts.**Not primary lift:** the rotors close the attitude-authority gap.**Propellers off:** attitude diverges and the robot falls. **Propellers on:** it remains upright and keeps hopping.

3. Platform - hardware built for leg-first hopping



Robot design

Leg and tri-rotor components

Electronic architecture

3-RSR parallel leg

Three RSR chains meet at one foot: fast 3-DoF omnidirectional placement, only **0.35 kg** moving mass.

Computational hub

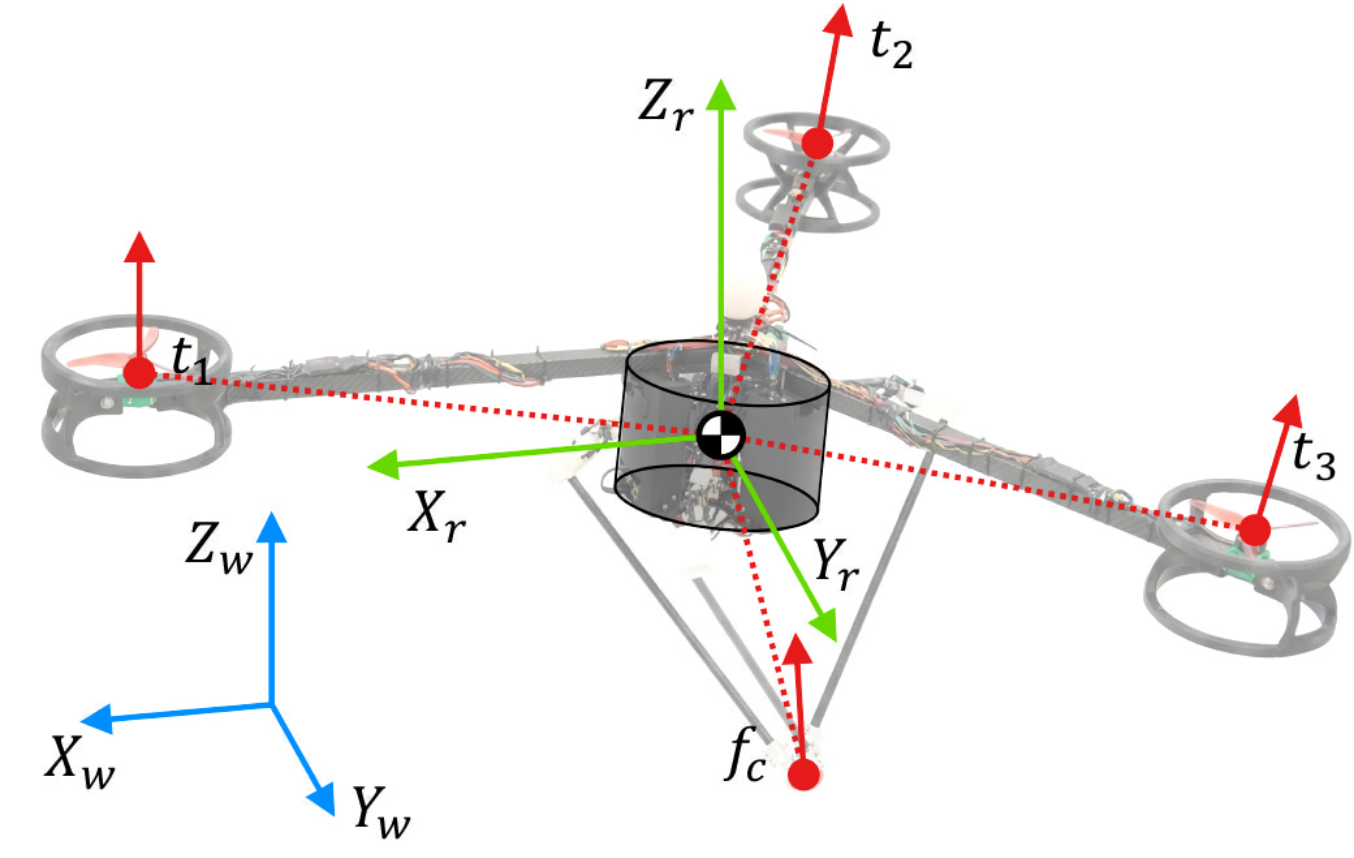
Batteries and electronics near the hip: low center of mass, low inertia.

Y-layout tri-rotor

Continuous attitude authority; **10 N** per rotor, total **0.82 mg**: a hopper, not a drone.

3.75 kg total ▪ 9.2% leg mass ▪ 20 Nm / leg motor ▪ 500 Hz control

4. Model - one rigid body, two actuator sets

Contact force f_c and thrusts $t_{1,2,3}$ produce the body wrench.

Net wrench = leg contact + three rotors

$$\begin{bmatrix} F_{net} \\ \tau_{net} \end{bmatrix} = \begin{bmatrix} f_c \\ r_c \times f_c \end{bmatrix} + \sum_{i=1}^3 \begin{bmatrix} d_i \\ r_i \times d_i \end{bmatrix} t_i$$

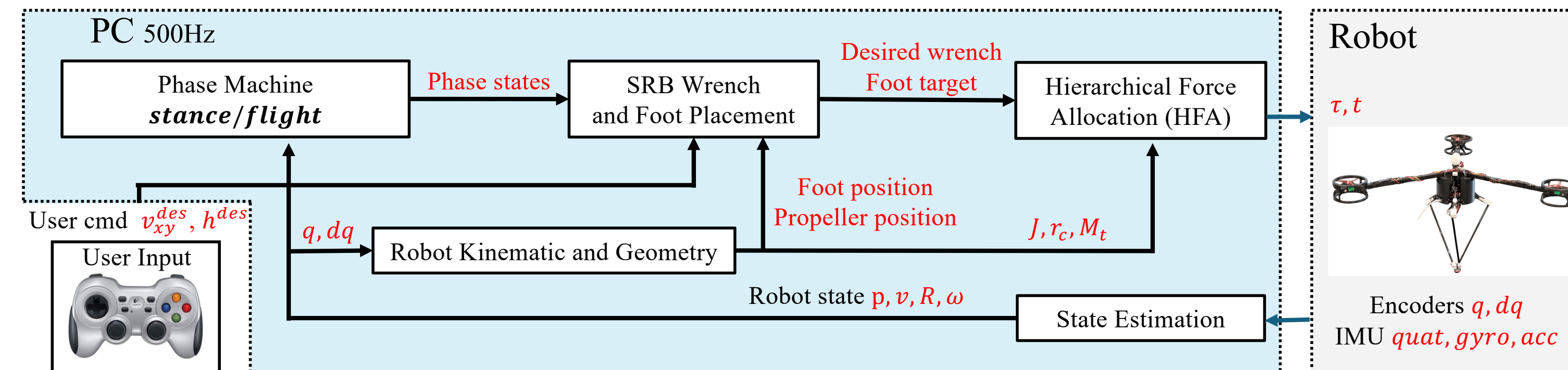
Newton-Euler dynamics of the CoM

$$m\dot{a} = F_{net} + mg, \quad \dot{\omega} = I^{-1}(R^T \tau_{net} - \omega \times I\omega)$$

Symbols (values used in all experiments)

 f_c ground reaction force (world) r_c, r_i arms: CoM → contact, rotor i $t_i \in [0, 10]$ N thrust along $d_i = Re_3$ F_{net}, τ_{net} net force, moment $m=3.75$ kg; g gravity; I inertia $R \in SO(3)$ attitude; ω body rate J leg Jacobian; $\tau_{max}=20$ Nm limit $\mu=0.4$ friction coefficient $p_z, \dot{p}_z$ CoM height, velocity; $h_{des}=0.664$ m l leg length; $l_0=0.464$ m; Δ_{ld} threshold k_z, b_z, k_E vertical, k_v, k_r Raibert gains K_R, K_ω attitude PD gains

5. Control - leg-priority hierarchical force allocation



500 Hz architecture: commands + feedback → planner → two-stage allocator → leg torques + rotor thrusts.

A. Vertical energy hopping (stance)

$$E_{sys} = \frac{1}{2} m \dot{p}_z^2 + mg p_z, \quad E_{des} = mg h_{des}$$

$$f_{c,z}^{des} = \max(0, k_z(h_{des} - p_z) - b_z \dot{p}_z + k_E(E_{des} - E_{sys}))$$

Injects push-off energy, absorbs impact.

B. SO(3) attitude PD (both phases)

$$e_R = \frac{1}{2} (R_{des}^T R - R^T R_{des})^V$$

$$\tau_0^{des} = -K_R e_R + K_\omega (\omega^{des} - \omega)$$

Roll/pitch only; yaw left free. $\tau^{des} = R \tau_0^{des}$.

C. Leg first - constrained least squares

$$f_{c,xy}^{cmd} = \arg \min_{f_{xy}} \|(r_c \times f_c)_{rp} - (\tau_{res}^{des})_{rp}\|_2^2$$

$$\text{s.t. } \|f_{xy}\|_2 \leq \mu f_{c,z}^{des}, \quad \|J^T(-R^T f_c)\|_\infty \leq \tau_{max}$$

The leg realizes all it can under friction and torque limits.

D. Rotors take only the residual

$$\tau_{res}^{cmd} = \tau^{des} - r_c \times f_c^{cmd}$$

$$\delta \tau_{res}^{cmd} = \arg \min_{\delta \tau} \|(M_I \delta \tau)_{rp} - (\tau_{res}^{cmd})_{rp}\|_2^2$$

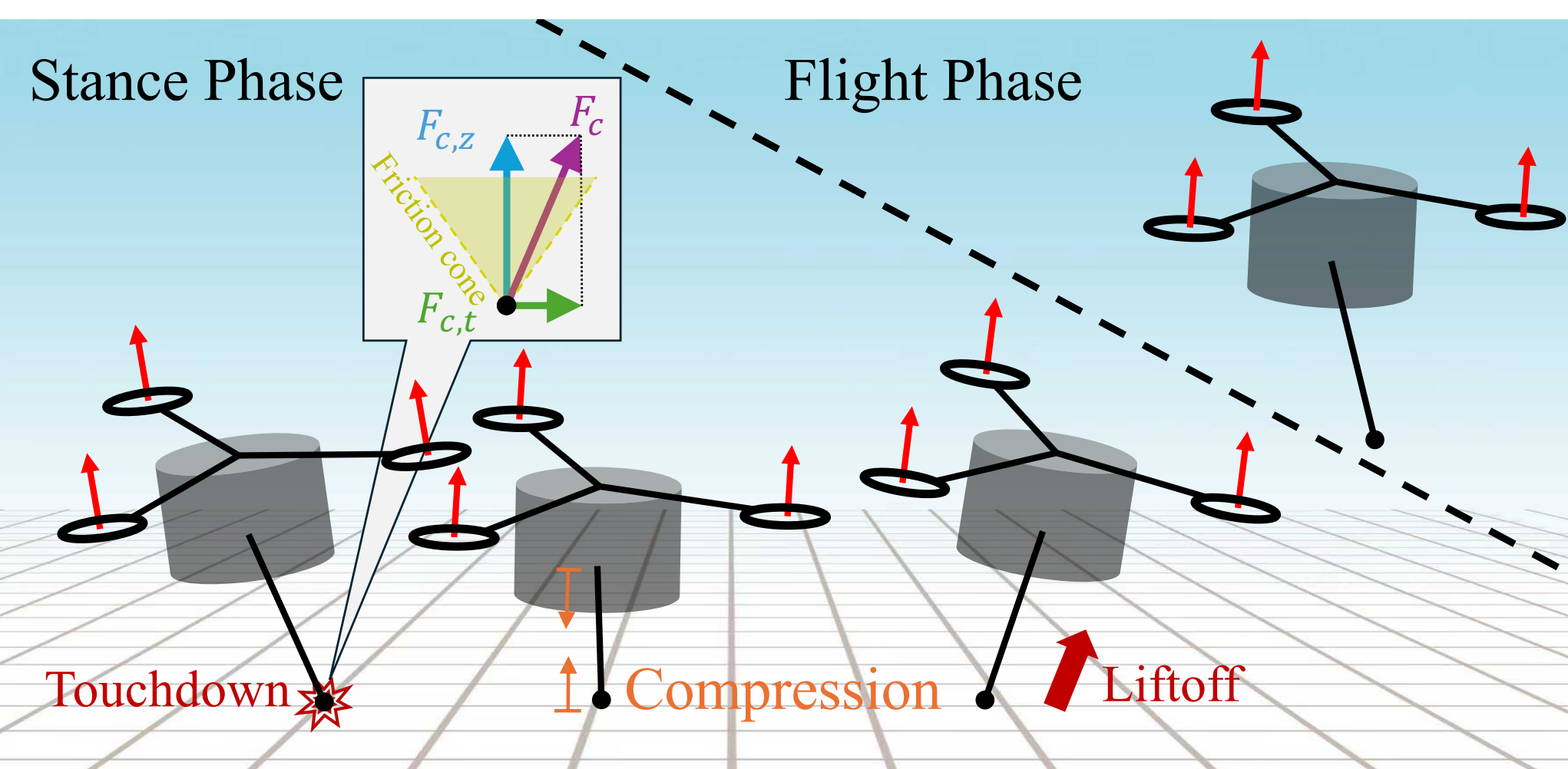
s.t. thrust bounds; $t^{cmd} = \frac{T_{base}}{3} \mathbf{1}_3 + \delta t^{cmd}$, $T_{base} = 0.02 mg$.

E. Flight - rotors primary for attitude; Raibert foot placement

$$f_c = 0; \quad \text{rotors track } (\tau^{des})_{rp}; \quad \text{leg tracks } r_{td,xy}^{des} = k_v v_{xy} - k_r v_{xy}^{des}$$

By construction the rotors can only add to - never replace - the leg's contact action.

6. Onehop - event-driven authority handoff



Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.

Touchdown: $l \leq l_0 - \Delta_{ld}$

Leg absorbs impact; energy law drives the next apex.

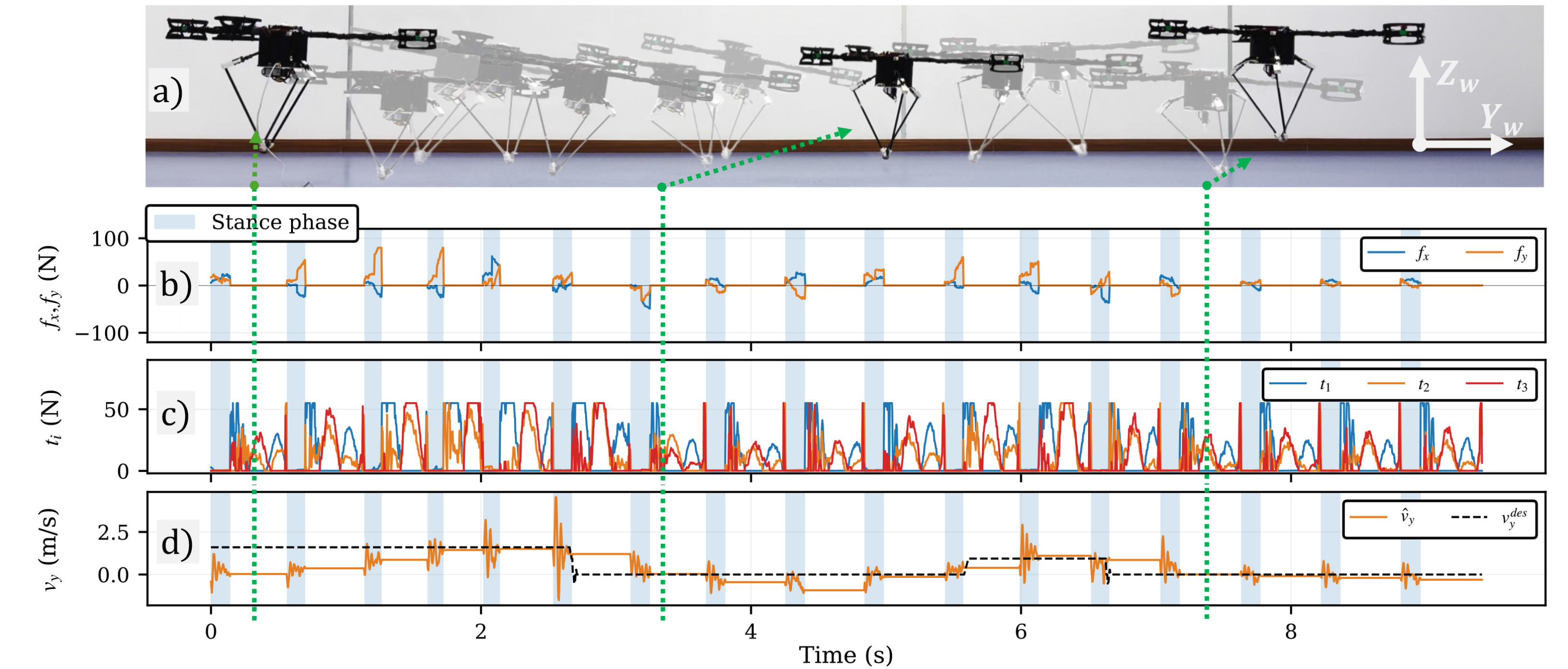
Liftoff: $l \geq l_0$

Contact wrench vanishes; rotors primary for attitude.

Handoff

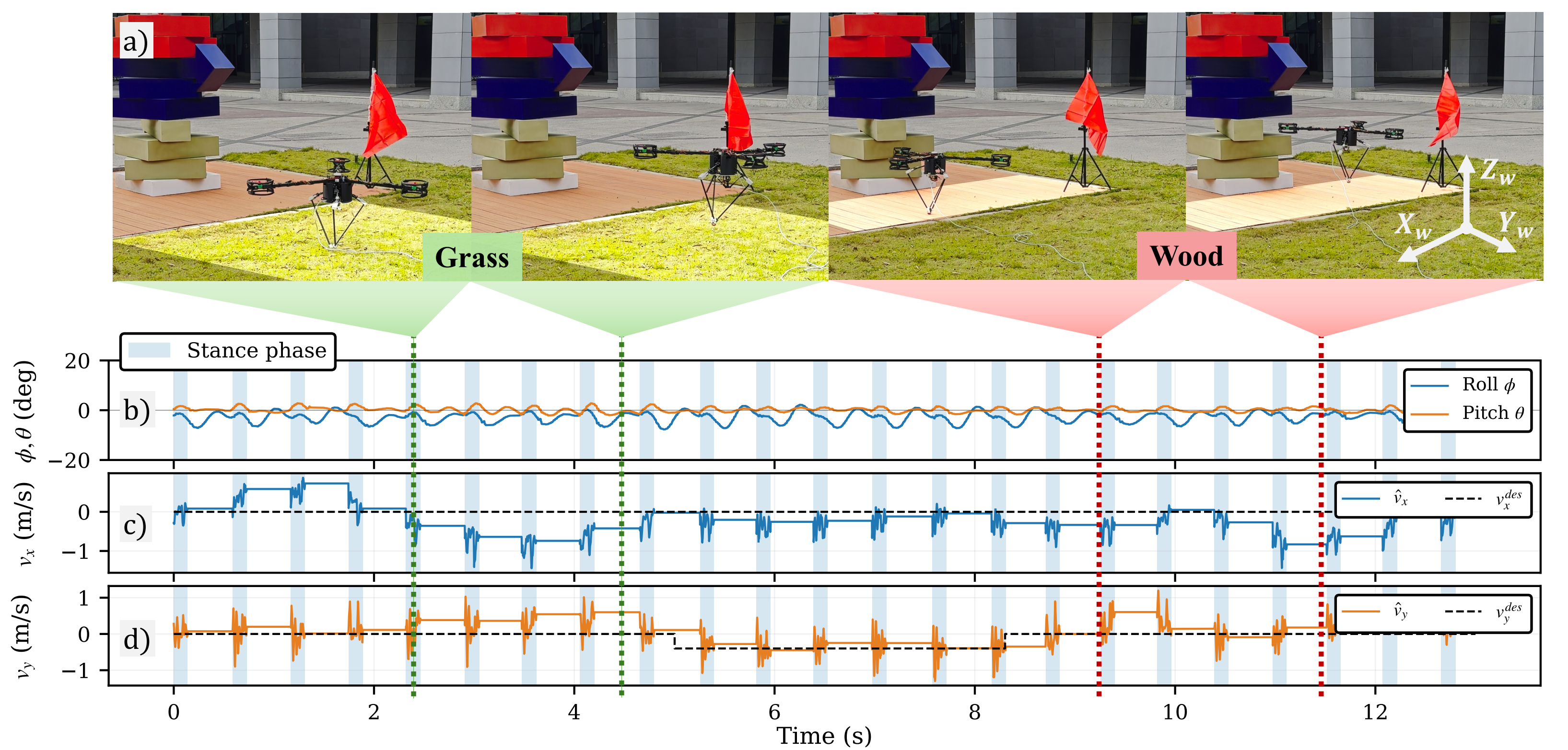
Events switch authority; no hardware or gain change.

7. Validation - indoor tracking



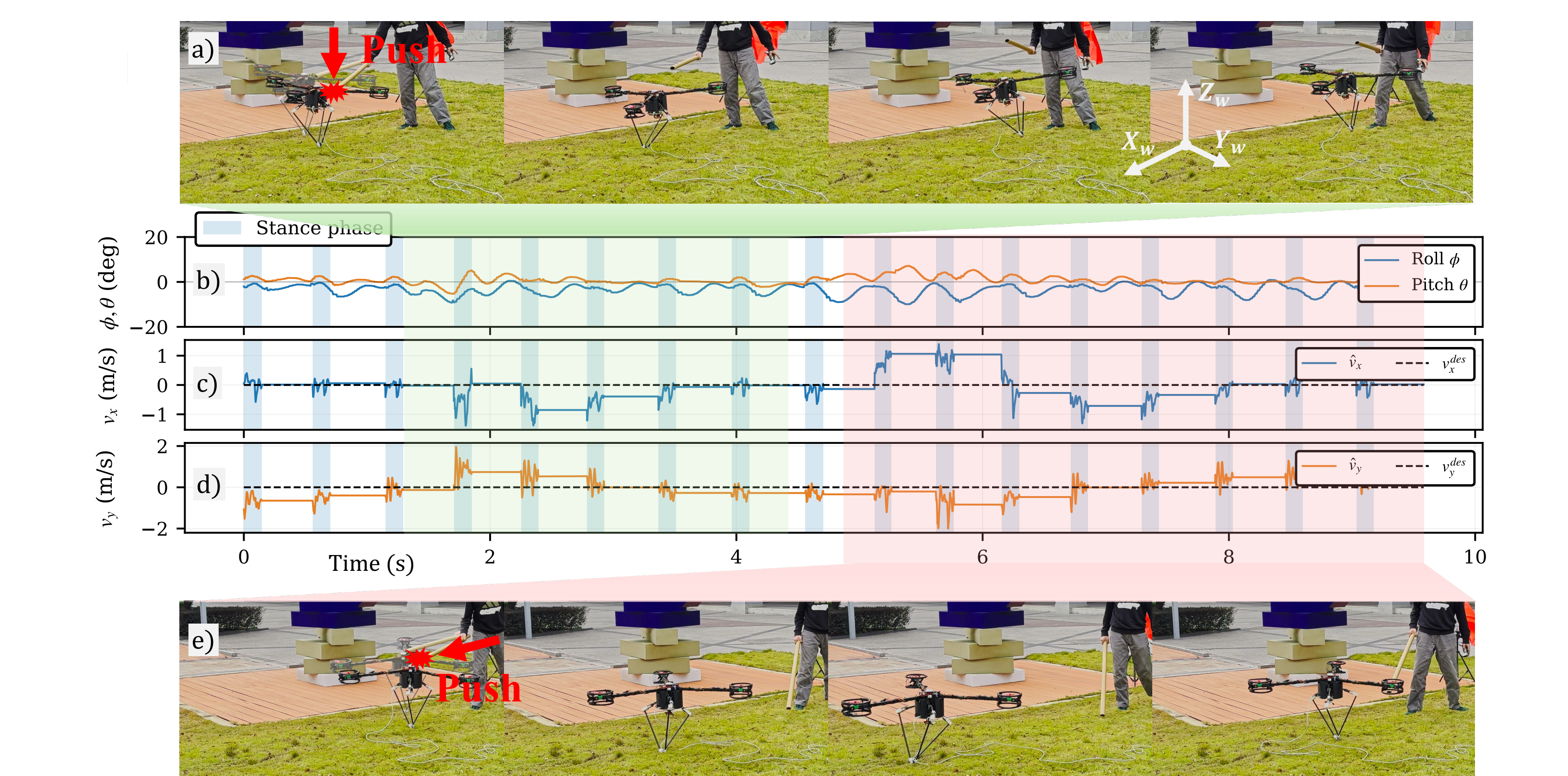
17 hops - 1.6 and 0.94 m/s commands - bounded attitude error through every impact.

8. Validation - grass to wood



23 hops - same controller - no terrain model - no gain retuning.

9. Validation - push recovery



Rotor-arm hit: attitude recovers in 1-2 hops. Main-body hit: lateral velocity recovers in three hops.